Internship Report – Day 13

Intern Name: [Your Name]
Internship Domain: Cloud Computing / DevOps
Organization: [Company/Institution Name]
Date: [Day 13 Date]
Duration: [e.g., 9:30 AM – 4:30 PM]

Objective of the Session

The objective of Day 13 was to gain a detailed understanding of **Azure Policy** for cloud governance and compliance, and to learn the fundamentals of **Docker**, including containerization concepts and commonly used Docker commands.

Topics Covered

1. Azure Policy (Detailed Learning)

Azure Policy is a governance tool in Microsoft Azure used to enforce organizational standards and assess compliance across cloud resources.

Key Concepts Learned

- Definition of Azure Policy and its importance in cloud governance
- Policy definitions and initiatives
- Scope of policies (Management Group, Subscription, Resource Group, Resource)
- Policy assignments
- Compliance evaluation and reporting
- Built-in vs Custom policies

Main Features

- Enforces rules automatically during resource creation
- Ensures compliance with organizational standards
- Helps control costs and security risks
- Supports auditing and remediation

Policy Effects Learned

- **Deny** – Prevents non-compliant resources from being created
- **Audit** – Flags non-compliant resources without blocking them
- **Append** – Adds additional configuration settings

- **DeployIfNotExists** – Deploys required configurations automatically
- **Modify** – Updates resource properties during deployment

Practical Understanding

- Creating and assigning policies in Azure Portal
- Monitoring compliance status
- Understanding real-world use cases like restricting regions and enforcing tagging standards

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2. Introduction to Docker

Docker is a containerization platform used to package applications along with their dependencies so they run consistently across different environments.

Concepts Learned

- Difference between Virtual Machines and Containers
- Docker Architecture
  - Docker Client
  - Docker Daemon
  - Docker Images
  - Docker Containers
  - Docker Hub
- Benefits of containerization:
  - Lightweight deployment
  - Faster startup
  - Environment consistency
  - Easy scalability

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3. Docker Commands (Hands-on Learning)

We practiced essential Docker commands used in real-world DevOps workflows.

Basic Docker Commands

- `docker --version`
  Checks Docker installation.
- `docker pull <image>`
  Downloads an image from Docker Hub.
- `docker images`
  Lists available images.
- `docker run <image>`
  Creates and runs a container.

- `docker ps`
  Shows running containers.
- `docker ps -a`
  Displays all containers (running and stopped).
- `docker stop <container_id>`
  Stops a running container.
- `docker start <container_id>`
  Starts a stopped container.
- `docker rm <container_id>`
  Removes a container.
- `docker rmi <image_id>`
  Deletes an image.

Additional Useful Commands

- `docker exec -it <container_id> bash` — Access container terminal
- `docker logs <container_id>` — View container logs
- `docker build -t <image_name> .` — Build image from Dockerfile

Skills Gained

- Understanding cloud governance using Azure Policy
- Managing compliance and enforcing rules in Azure
- Basic containerization knowledge using Docker
- Running and managing containers through CLI commands
- Understanding DevOps deployment workflows

Learning Outcome

By the end of the session, I understood how Azure Policy helps organizations maintain security and compliance in cloud environments. Additionally, I gained practical exposure to Docker and learned how containers simplify application deployment and environment management.

Conclusion

Day 13 provided valuable insights into cloud governance and container technology. Azure Policy demonstrated how organizations control and standardize cloud resources, while Docker introduced modern application deployment practices widely used in DevOps and cloud-native environments.